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Chapter 1

Stage A Molnar Single-Notch Benchmark

1.1 Objective and Decision Boundary

Stage A establishes whether the Molnar and Gravouil single-notch phase-field benchmark is a defensible baseline for later Abaqus adaptive-remeshing work. The aim is deliberately narrower than the full thesis workflow: reproduce an established staggered phase-field implementation, quantify the reconstructed single-notch response, and decide whether the evidence is strong enough to authorize Stage B uniform-reference studies.

The current Stage A status is:

```
candidate v2: technically passed
scientific review: incomplete
SDV15: completed-increment irreversibility violation
Gate A3: reference_data_insufficient
new solution run: not authorized
```

This chapter therefore treats candidate v2 as a technically reproducible and useful reproduction baseline, but not as a Gate A3 pass. Supervisor approval is required before any Stage B, uniform-reference, MISESeri, remeshing, or state transfer work begins.

1.2 Benchmark Choice

The Molnar single-notch benchmark [?] was selected because it connects directly to the thesis requirement for an Abaqus UEL/UMAT phase-field workflow. Related Abaqus phase-field UEL architectures [?] and the local multi-field implementation context [?] inform the interpretation of interfaces and visualization layers, but do not replace the Molnar staggered baseline. The source package contains a staggered displacement/phase implementation, a visualization layer through state variables, and single-notch examples suitable for checking reaction force, crack-path geometry, and phase-field/history behavior before introducing adaptive remeshing.

The benchmark is also a useful stress test for thesis discipline. It separates technical execution from scientific validation: a model can compile, solve, and produce a plausible crack while still failing a strict scientific gate because the RF-U reference, tolerances, or irreversibility interpretation remain unresolved.

1.3 Formulation Convention

Foundational phase-field fracture formulations [?, ?] motivate the regularized crack description used here, while the implemented staggered solution follows Molnar and Gravouil [?]. The phase-field convention used throughout Stage A is:

- $d = 0$: intact material.
- $d = 1$: fully broken material.

Candidate v2 preserves the published Molnar source formulation. The generated source copy `models/generated/molnar_gravouil_2017/paper_matched_single_notch_v2/SingleNotch_v2.for` is copied from the preserved source with only the physical-element count `N_ELEM=33852` updated for the reconstructed mesh. The original source is preserved at `models/baseline_original/molnar_gravouil_2017/02_Single_Notch_Tension/SingleNotch.for`.

1.4 Staggered Phase/Displacement Method

The implementation follows a staggered solution structure with displacement and phase-field layers. The displacement field is solved in the mechanical layer, and the phase-field layer updates the scalar damage variable using the stored history information. Visualization variables are copied into a companion UMAT/CPS4 layer; those visualization fields are evidence outputs, not separate governing fields.

For thesis use, three distinctions must remain visible:

- The published formulation is the Molnar and Gravouil staggered phase-field UEL/UMAT implementation.
- The reconstructed implementation is the project candidate-v2 paper-matched deck generated from documented geometry, loading, material, and mesh assumptions.
- The approximate Fig. 7 reference is a local digitization of the published curve, not exact author data and not a supervisor-approved final tolerance basis.

1.5 Geometry, Material, and Loading

Candidate v2 reconstructs a $1.0 \text{ mm} \times 1.0 \text{ mm}$ single-edge-notched plate with a left-edge split-node notch of length 0.5 mm at $y = 0$. The model uses plane strain and unit thickness. Material parameters are $E = 210 \text{ kN/mm}^2$, $\nu = 0.3$, $G_c = 0.0027 \text{ kN/mm}$, and $l_c = 0.0075 \text{ mm}$.

The load angle is 90° . The loading is applied in two direct-increment steps:

Step	Start U_2 (mm)	End U_2 (mm)	Increment (mm)
Step 1	0.0000	0.0050	1.0×10^{-5}
Step 2	0.0050	0.0067	1.0×10^{-5}

1.6 Paper-Matched Reconstruction

The reconstruction is documented in the project configuration and derived reference records. Candidate v2 restores the intended loading arithmetic, implements a split-node notch, preserves the layered UEL/UMAT architecture, and uses the source property layout from the preserved Molnar files. Static validation passed, and the mesh-quality preflight classified the model as `mesh_quality_preflight_pass`.

The mesh has 33852 physical elements and 101556 layered elements. The local mesh size is $h = 0.001$ mm, giving $h/l = 0.1333333333$. The maximum element size is 0.025 mm, the maximum neighboring-size ratio is 1.5, and the maximum aspect ratio is 25.0000000000001368. The high-aspect-ratio elements are documented as reconstruction limitations outside the fracture-process corridor.

1.7 Technical Validation

The single authorized serial HPC baseline job was 1374864.mm`master02`, submitted from revision 711dd495bdc830d695f9d7e56283316c9d417d5. It completed with PBS Exit_status=0; Abaqus returned code zero; the .sta, .msg, .dat, and scratch .odb outputs existed; and the .sta reported successful analysis completion. The technical classification is paper_matched_v2_technical_pass

The run used one CPU, 32 GB requested memory, and a 24-hour walltime request. Recorded resource use was walltime 00:38:38, CPU time 00:35:52, CPU percent 95, memory 2970760 kb, virtual memory 3565868 kb, and an Abaqus-reported peak memory of about 2 GB.

1.8 Numerical Results

Table 1.1: Stage A candidate-v2 numerical evidence. Values are drawn from committed extraction, scientific-review, and targeted-diagnostic evidence.

Metric	Value	Evidence
Peak RF2	0.761702 kN	runs/hpc/paper_matched_single_notch_v2/scientific_review/fig7_comparison_metrics.json
Peak displacement	$U_2 = 0.006110$ mm	runs/hpc/paper_matched_single_notch_v2/scientific_review/fig7_comparison_metrics.json
Peak-force error	0.064519 relative	runs/hpc/paper_matched_single_notch_v2/scientific_review/FIG7_COMPARISON_AUDIT.md
Peak-displacement error	0.041257 relative	runs/hpc/paper_matched_single_notch_v2/scientific_review/FIG7_COMPARISON_AUDIT.md
Pre-peak RMSE	0.044136 kN	runs/hpc/paper_matched_single_notch_v2/scientific_review/FIG7_COMPARISON_AUDIT.md

Metric	Value	Evidence
Post-peak RMSE	0.348093 kN	runs/hpc/paper_ matched_single_notch_ v2/scientific_review/ FIG7_COMPARISON_ AUDIT.md
Full-overlap NRMSE	0.245705	runs/hpc/paper_ matched_single_notch_ v2/scientific_review/ fig7_comparison_ metrics.json
Tangent-stiffness error	about +51.4%	runs/hpc/paper_ matched_single_notch_ v2/scientific_review/ SCIENTIFIC_DECISION. md
Common-domain area error	+0.193324 relative	runs/hpc/paper_ matched_single_notch_ v2/scientific_review/ fig7_comparison_ metrics.json
Crack extension, $SDV15 \geq 0.80$	0.0555 mm beyond notch tip; 0.0570 mm total connected path	runs/hpc/paper_ matched_single_notch_ v2/scientific_review/ SCIENTIFIC_DECISION. md
Crack extension, $SDV15 \geq 0.90$	0.0525 mm beyond notch tip; 0.0530 mm total connected path	runs/hpc/paper_ matched_single_notch_ v2/scientific_review/ SCIENTIFIC_DECISION. md
Crack extension, $SDV15 \geq 0.95$	0.0505 mm beyond notch tip; 0.0510 mm total connected path	runs/hpc/paper_ matched_single_notch_ v2/scientific_review/ SCIENTIFIC_DECISION. md
Crack extension, $SDV15 \geq 0.99$	0.0465 mm beyond notch tip; 0.0460 mm total connected path	runs/hpc/paper_ matched_single_notch_ v2/scientific_review/ SCIENTIFIC_DECISION. md
SDV15 maximum overshoot	1.005600	runs/hpc/paper_ matched_single_notch_ v2/scientific_review/ sdv_irreversibility_ metrics.json
Accepted-increment decreasing transi- tions	2184	runs/hpc/paper_ matched_single_notch_ v2_sdv15_diagnostic_ r2/RUN_SUMMARY.md

Metric	Value	Evidence
Transitions above 10^{-6}	2131	runs/hpc/paper_ matched_single_notch_ v2_sdv15_diagnostic_ r2/evidence/ 1375028.mmaster02/ postprocessing_ completed_increment_ replay_time_aligned/ severity_audit/sdv15_ completed_increment_ severity_summary.json
Transitions above 10^{-5}	1362	runs/hpc/paper_ matched_single_notch_ v2_sdv15_diagnostic_ r2/evidence/ 1375028.mmaster02/ postprocessing_ completed_increment_ replay_time_aligned/ severity_audit/sdv15_ completed_increment_ severity_summary.json
Maximum decrease	$2.2384088238425193 \times 10^{-4}$	runs/hpc/paper_ matched_single_notch_ v2_sdv15_diagnostic_ r2/RUN_SUMMARY.md
Mean decrease	$2.8434597987446906 \times 10^{-5}$	runs/hpc/paper_ matched_single_notch_ v2_sdv15_diagnostic_ r2/RUN_SUMMARY.md
Median decrease	$1.3940791172561973 \times 10^{-5}$	runs/hpc/paper_ matched_single_notch_ v2_sdv15_diagnostic_ r2/RUN_SUMMARY.md
Unique affected element/IP locations	138	runs/hpc/paper_ matched_single_notch_ v2_sdv15_diagnostic_ r2/RUN_SUMMARY.md
SDV16 decreases	0	runs/hpc/paper_ matched_single_notch_ v2_sdv15_diagnostic_ r2/RUN_SUMMARY.md
Diagnostic RF–U perturbation	maximum normalized RF2 difference $6.54442468760855 \times 10^{-13}$; maximum U2 difference $4.992967844730245 \times 10^{-15}$ mm	runs/hpc/paper_ matched_single_notch_ v2_sdv15_diagnostic_ r2/RUN_SUMMARY.md

Metric	Value	Evidence
Solver walltime and memory	baseline PBS wall-time 00:38:38, memory 2970760 kb, Abaqus peak memory about 2 GB	<code>runs/hpc/paper_matched_single_notch_v2/scientific_review/solver_resource_metrics.json</code>

1.9 RF–U Interpretation

The RF–U comparison is mixed. The peak force scale is close enough to justify continued review, but the full curve is not within a strict provisional working gate. The pre-reference-peak RMSE is 0.044136 kN, while the post-reference-peak RMSE is 0.348093 kN. This makes the post-peak mismatch the dominant scientific limitation. The comparison uses the approximate digitized Fig. 7 red dashed $l_c = 0.0075$ mm reference, which has estimated digitization uncertainty of about ± 0.00004 mm and ± 0.015 kN.

1.10 Crack-Path Interpretation

The final high-damage band is connected and approximately horizontal at thresholds from $\text{SDV15} \geq 0.80$ through $\text{SDV15} \geq 0.99$. At the provisional $\text{SDV15} \geq 0.95$ threshold, the final matched contour contains 193 element-mean damaged elements and a connected high-damage extension of about 0.0505 mm beyond the notch tip. This supports qualitative crack-direction agreement, but it remains threshold-dependent and does not close Gate A3 by itself.

1.11 SDV15, SDV16, and Irreversibility

In the project convention, SDV15 is the phase-field visualization variable and SDV16 is the history-like variable used to monitor monotonic driving behavior. The original ODB scan found SDV16 monotone and SDV15 nonmonotone in retained visualization frames. A later targeted diagnostic r2 run was non-intrusive by RF–U comparison and enabled a no-solution completed-increment replay of the existing trace rows.

The completed-increment replay kept the last U1 stage-101 call for each (KSTEP, KINC, physical element, so. It processed 627304 trace rows, retained 209152 U1 stage-101 rows, formed 101840 final increment states over 152 element/IP sequences, and found 2184 unique completed-increment SDV15 decreasing transitions. The severity audit classifies this as `sdv15_completed_increment_irreversibility_vio` under the provisional 10^{-6} materiality tolerance. SDV16 has zero decreases over the same final-increment sequences.

1.12 Scientific Limitations

The current evidence supports the following restrained interpretation:

- Candidate v2 is technically reproducible.
- Crack propagation direction is qualitatively correct.
- RF–U agreement is mixed, and post-peak disagreement is substantial.
- SDV16 remains monotone in the checked sequences.
- SDV15 decreases between accepted increments.

- The targeted diagnostic instrumentation is non-intrusive.
- Strict pointwise phase-field irreversibility is not satisfied.
- Candidate v2 remains a useful reproduction baseline for supervisor review.
- Supervisor approval is required before Stage B or any downstream adaptive-remeshing work.

1.13 Supplementary $l_c = 0.015$ mm mesh h-convergence (RF–U)

For the exact author-supplied supplementary single-notch model ($l_c = 0.015$ mm), three serial meshes were solved and postprocessed for RF–U only. Successive peak-force changes are approximately 4.0% (H0→H1) and 0.47% (H1→H2-PUB); pre-peak curve NRMSE between H1 and H2-PUB is about 0.11%, while post-peak residual remains larger ($\sim 20\%$ NRMSE). After supervisor Decisions **1A** and **2B**, mesh roles are frozen as: H2-PUB for fine RF–U validation only; H1 for production/thesis/report simulations and the Stage C comparison reference; H0 for fast testing and the first MISESeri implementation trials. Crack-path/contour mesh convergence remains deferred (Decision **2B**) and does not block Stage C preparation. Evidence: `runs/hpc/molnar_lc015_h_convergence/comparison/H_CONVERGENCE_SCIENTIFIC_REVIEW.md`; `docs/decisions/MOLNAR_GATE_A3_SUPERVISOR_DECISION_1A_2B.md`; `docs/decisions/MESH_USE_POLICY.md`.

1.14 Gate A3 Boundary

Gate A3 is **conditionally accepted for RF–U validation use** after Decisions **1A** and **2B** (`gate_a3_conditionally_accepted_rf_u`; `contour_validation_deferred`). Full unconditional closure of every historical Stage A claim is not asserted; residual items may still retain a `reference_data_insufficient` narrative where appropriate. Candidate v2 has not been reclassified as a full scientific pass by these decisions. Stage C MISESeri **preparation** is authorized; HPC submission, MISESeri execution, multicore qualification, and state-transfer work remain unauthorized without explicit further approval. Stage C preparation plan: `docs/studies/STAGE_C_MISESERI_PREPARATION_PLAN.md`.

Chapter 2

Offline MISESeri Pre-Refinement for Layered Phase-Field UEL Models

2.1 Objective

Demonstrate a custom, auxiliary-continuum, MISESeri-driven offline pre-refinement workflow for the Molnar layered phase-field UEL model [?], targeting the elastic, pre-peak and peak RF–U response of uniform H1 with fewer physical elements. The indicator-driven pre-refinement strategy is motivated by published Abaqus MISESeri practice for phase-field fracture [?] and by the product documentation for error-indicator-based mesh refinement [?]; the present Stage C result is a scoped offline demonstration, not online adaptive remeshing.

2.2 Method summary

1. Solve an auxiliary continuum pre-analysis with active MISESeri output.

2. Mark elements with

$$\frac{\text{MISESeri}}{\max(\text{MISESeri})} > \text{relative_MISESeri_threshold}, \quad \text{relative_MISESeri_threshold} = 0.05.$$

3. Build a localized graded physical mesh (min size 0.0025 mm, max size 0.025 mm, one pass, no coarsening).

4. Reconstruct U1/U2/CPS4 layers and Fortran N.ELEM.

5. Solve with four OpenMP threads and compare to uniform H1.

2.3 Verification narrative

A failed iteration (C2F-v2) omitted the exact $y = 0$ mesh line, so notch free faces were not reconstructed. The resulting continuous plate elevated initial stiffness by about 72% and suppressed softening. Geometry audit, restoration of $y = 0$, elastic probe (0.083% stiffness error), phase-field spatial check, and C2F-v3 restored peak/pre-peak agreement with H1.

2.4 Results (C2F-v3)

Peak force difference 0.24%; initial stiffness difference 0.061%; pre-peak NRMSE 0.089%; peak displacement identical at 0.0058 mm; 10290 physical elements (14.7% fewer than H1). A direct repeat solve, job 1376595.mm.master02, reproduced the frozen C2F-v3 RF–U metrics exactly at

the reporting precision: peak-force relative difference, pre-peak NRMSE, and initial-stiffness relative difference were all zero against the original job `1376480.mmaster02`. The final SDV15 field summary was also unchanged, with 41160 finite samples, 2247 samples above 0.5, and 464 samples above 0.95.

The fair four-thread comparison is now available from the H1 closeout job `1376594.mmaster02`. Both H1 and C2F-v3 used four OpenMP threads. Relative to the four-thread H1 baseline, C2F-v3 used 14.7% fewer physical elements, 16.7% less walltime, 14.9% less CPU time, and 22.2% less peak memory. These cost reductions support the efficiency demonstration, but they should not be attributed solely to remeshing without controlling for all queue, node, module, and output settings.

Post-peak RF–U equivalence remains limited: the post-peak NRMSE is about 24%. The matched-state SDV15 extraction and crack-path metrics jobs `1376596.mmaster02` and `1376597.mmaster02` completed successfully. They support C2F-v3 repeatability, but the quantitative H1-versus-refined crack path comparison is classified as `crack_path_deviation_detected`. At $U = 0.007$ mm and threshold $\text{SDV15} \geq 0.5$, H1 reaches an extension of 0.5075 mm whereas C2F-v3 reaches 0.4949 mm; the Hausdorff distance is about 0.0122 mm. Therefore the refined model is supported for elastic, pre-peak, and peak RF–U response, but not as a crack-path-equivalent replacement for H1.

2.5 Production policy

Uniform H1 remains the production and report reference. The refined mesh is a demonstration of offline pre-refinement efficiency, not a replacement of H1. Fair walltime comparisons should use the four-thread H1 closeout package and the frozen C2F-v3 job records. The first automation smoke job `1376598.mmaster02` completed with scheduler `Exit_status=12`; it is retained as failed guard evidence and is not used as a scientific result. The corrected T5 automation smoke rerun `1376758.mmaster02` completed with scheduler `Exit_status=0` and classification `automation_smoke_pass`. It generated and statically validated the 0.45 mm H0 notch variation with 1600 physical elements, preserved the exact $y = 0$ line and split notch faces, and executed a short elastic probe with $U = 9.9999997 \times 10^{-5}$ mm, $RF_2 = 0.0147068165243$, and $K_{\text{probe}} = 147.0681696550451$. This is workflow automation evidence only, not a fracture-reference comparison.

Chapter 3

Controlled State Transfer for Phase-Field UEL Workflows

3.1 Objective

Stage D tests whether phase-field and history values transferred between meshes are actually ingested by Abaqus and routed through the UEL/UMAT state arrays. The test is intentionally separated from fracture evolution. No interrupted Molnar fracture simulation is allowed until the D2 ingestion, continuation, thread-repeatability, and output-routing gates pass.

3.2 D1 analytical baseline

The D1 harness established deterministic and physically guarded transfer mechanics, but it did not prove negligible transfer error. The retained D1 baselines are:

Quantity	Value
Nodal d L2 error	0.0270
Nodal d maximum error	0.0850
Integration-point H L2 error	0.0108
Integration-point H maximum error	0.0234
Energy difference	−0.00769

3.3 D2A ingestion design

D2A uses the prepared tiny nonmatching transfer package under `models/state_transfer/d2_tiny_transfer/`. A separate D2-only UEL/UMAT source variant is generated under `src/state_transfer/d2_tiny_transfer_ue`. The frozen Molnar source and accepted C2C-v3 mesh are not modified.

The preserved Molnar U1 user-element card declares phase on Abaqus DOF 3. D2A therefore prescribes every transferred nodal phase value on DOF 3. History values are loaded from a generated read-only table keyed by physical element label and integration point number, initialized once, and mirrored to visualization SDV16. The interpolated phase value is mirrored to SDV15.

3.4 Current status

The executable D2A package passed on HPC job 1376785.mmaster02. The PBS exit status was zero, the Abaqus solve completed successfully, the ODB was readable, and D2A.ok was written.

Target element/IP coverage was 1.0, the maximum SDV15 interpolation error was 0.0, and the maximum SDV16/ H error was $6.42899999030793 \times 10^{-9}$.

Abaqus did not expose the UEL phase DOF as usable nodal U output in this smoke deck. The accepted D2A proof is therefore the element/IP UMAT mirror: SDV15 proves transferred phase ingestion, and SDV16 proves transferred history ingestion. This is an ingestion and routing verification, not a fracture-response benchmark.

3.5 D2B continuation attempt

D2B was submitted once as serial PBS job 1376819.mmamster02. It did not pass. Abaqus compiled the D2-only user subroutine and entered the three-step deck, but the tiny mechanical continuation terminated with solver exit 1 and PBS `Exit_status=10`. The message file reports `TOO MANY INCREMENTS NEEDED TO COMPLETE THE STEP`. Therefore no D2B.ok marker exists, and no serial state-persistence acceptance claim is made from this attempt.

The prepared corrective deck increases the release and continuation step increment allowance to `inc=50`. It leaves the transferred phase values, history values, transfer table, label mapping, and D2-only UEL/UMAT state route unchanged.

3.6 D2B accepted rerun

The corrected D2B rerun, 1376825.mmamster02, passed with PBS exit status zero, solver exit zero, readable ODB output, and classification `stage_d2b_serial_continuation_pass`. The target element/IP coverage was 1.0. The maximum initial differences from accepted D2A were 0.0 for both SDV15 and SDV16; the maximum release differences were also 0.0. The final continuation frame reached $U_2 = 1.0 \times 10^{-5}$ mm with finite $RF2 = 3.46317381026 \times 10^{-7}$. The reported ALLWK continuation jump was $1.2830926425511091 \times 10^{-11}$; no energy-jump acceptance tolerance is assigned at D2B.

The rerun wrote D2B.R1.ok, and the canonical D2B.ok marker records the accepted rerun job ID.

3.7 D2C four-thread repeatability

The D2C job, 1376831.mmamster02, reused the accepted D2B transfer deck with no physics, transfer-table, mesh, label, step, increment-control, or target displacement changes. The only execution change was from one serial thread to `cpus=4, mp_mode=threads`, and four OpenMP threads. The Abaqus message output confirmed `1 MPI RANK x 4 THREAD`.

D2C passed with PBS exit status zero, solver exit zero, readable ODB output, and classification `stage_d2c_thread_repeatability_pass`. Against the D2B serial reference, the maximum SDV15 and SDV16 thread-serial differences were both 0.0, target element/IP coverage was 1.0, the final U_2 difference was 0.0 mm, the final RF2 absolute and relative differences were both 0.0, the RF-U NRMSE was 0.0, and the final-frame ALLWK absolute difference was 0.0. The accepted and cutback increment sequence was unchanged.

D2D ABAQUSER output-route verification is the next controlled gate.

3.8 D2D0 ABAQUSER availability

The D2D0 login-node audit did not find an ABAQUSER executable, module, source implementation, or documented runnable interface. The audit therefore classified D2D as `stage_d2d_blocked_abaquser_no`. This is an external-tool availability block, not a failure of D1-D2C state transfer. A normal Abaqus Python extractor is not an acceptable substitute for ABAQUSER verification.

The first interrupted fracture-transfer experiment has been prepared only as a design, using a small H0 diagnostic model and an early pre-peak checkpoint near $U = 0.003$ mm. No D2D PBS job or interrupted fracture solver job has been submitted.

3.9 D3A existing H0 checkpoint audit

The existing accepted H0 ODB from job 1376154.mmaste02 was tested as the D3 checkpoint source. A CAE/ODB-only extraction selected the frame at $U_2 = 0.003000000026077032$ mm and extracted 15720 element/IP state rows with full coverage, finite SDV15/SDV16, maximum $d = 0.08412302285432816$, maximum $H = 0.0512588769197464$, checkpoint RF2 = 0.39450356364250183, and $RF2/RF2_{peak,H0} = 0.5421925638518931$.

The source ODB lacks ALLIE, ALLSE, and ALLWK global energy history outputs, so D3A required independent energy evidence before acceptance. That original blocker is preserved in the history and resolved by the D3A-E R1 reconstruction.

3.10 D3A-E independent energy reconstruction

D3A-E replaced unavailable global ODB energy history with an independent quadrature reconstruction from the accepted H0 checkpoint exports. R0 job 1376885.mmaste02 is preserved as failed evidence with classification `stage_d3a_energy_reconstruction_fail_parser_scope`; the first parser mixed Part-1 mesh nodes and assembly reference-point nodes into one namespace.

R1 corrected the parser scope and passed with classification `stage_d3a_energy_reconstruction_pass`: 3930 physical elements, 15720 integration points, non-positive detJ count 0, minimum detJ $2.829135024804933 \times 10^{-6}$, and relative energy residual 0.012586306767288707. D3A is now closed as `stage_d3a_checkpoint_pass_independent_energy_reconstruction`; D3A2 nonmatching package construction is the next local gate.

3.11 D3A2 nonmatching package

D3A2 built and validated one deterministic nonmatching split-notch target package locally, without PBS submission. The accepted package classification is `stage_d3a2_transfer_package_pass`. The source is job 1376154.mmaste02 at checkpoint $U_2 = 0.003000000026077032$ mm.

The target has 6601 nodes, 6400 Q4 physical elements, and 25600 integration points. Its open notch runs from $x = -0.5$ to $x = 0.0$ on $y = 0$, with 40 duplicated coincident face-node pairs and a shared notch tip. The topology audit found notch length 0.5 mm, zero crossing elements, and zero non-positive Jacobian determinants. The transfer package gives target-node and target-IP coverage 1.0, predicted energy relative jump 0.015379624558651227, and unmapped state count 0. The package marker records `solver_job_submitted=false`; D3A3 ingestion/equilibration/release hold is the next gate before any continuation solve.

3.12 D3A3 ingestion hold attempt

The first D3A3 full-target ingestion/equilibration/release-hold job, 1377382.mmaste02, was submitted exactly once from the committed D3A2 package after static preflight passed. The job failed before solver analysis: Abaqus checked out a Standard license, but user-subroutine compilation could not find `ifort` because the batch script loaded only `abacus/2023`. The preserved classification is `stage_d3a3_solver_fail_compiler_environment`; no D3A3.ok marker exists and continuation stages D3D and D3E remain blocked.

D3A3-R1 applied this compiler-environment correction as job 1377383.mmaste02. The environment gate passed: the job recorded gcc/11.4.0, intel/2024.2.0, abaqus/2023, and a valid ifort. The analysis still did not start because the generated d3_transfer_table.inc exceeded the Intel Fortran statement token limit during user-subroutine compilation. The classification is stage_d3a3_solver_fail_transfer_table_compile; no D3A3_R1.ok or D3A3.ok marker exists, so D3D and D3E remain blocked.

D3A3-R2 is prepared as a compile/datacheck gate only. It keeps the R0 and R1 evidence directories unchanged and moves the new compile evidence to runs/hpc/stage_d3/interrupted_transfer/target_ingestion_compile_r2/. The oversized compile-time transfer include is replaced by a headerless runtime d3_transfer.h.dat file loaded once through UEXTERNALDB, while the physical Molnar H0 UEL/UMAT source remains the base implementation. Local runtime-state validation passed for 25600 H records, with minimum H $1.0045788889553414 \times 10^{-8}$, maximum H 0.027317036782803523, and SHA256 4689ea5c10c0972e69ba46f8676a326c8b011b98faa8031c7c26cfb218607cd9. The R2 static validation classification is stage_d3a3_static_validation_pass. The compile/datacheck job 1377389.mmaste02 compiled and linked the user subroutine and completed input processing, showing that the R1 compile-token problem was removed. Standard datacheck then failed because UEXTERNALDB opened relative file d3_transfer.h.dat from Abaqus' internal /local/... work directory where the file had not been staged. The classification is stage_d3a3_r2_datacheck_fail_runtime_h_file_not_in_abaqus_workdir. No D3A3_R2_COMPILE.ok marker exists, no full D3A3-R2 solver job has been submitted, and D3D/D3E remain blocked.

The bounded D3A3-R2-R1 correction then ran as job 1377391.mmaste02. It changed only the runtime-H file location logic: UEXTERNALDB calls GETOUTDIR, constructs OUTDIR/'/d3_transfer.h.dat', prints the resolved path, and opens FILE=HFILE(1:LHFILE). The path audit passed and the runtime-H SHA256 remained 4689ea5c10c0972e69ba46f8676a326c8b011b98faa8031c7c26cfb218607cd9. The prior Abaqus work-directory missing-file failure was removed: Standard reported the staged cluster runtime history path for d3_transfer.h.dat. Datacheck then failed with Intel Fortran severe (24), end-of-file during read on unit 99. The classification is stage_d3a3_r2_r1_datacheck_fail_runtime_h_eof_after_getoutdir_open. No D3A3_R2_COMPILE.ok marker exists, no full D3A3-R2 solver job has been submitted, and D3D/D3E remain blocked.

D3A3-R2-R2 is prepared as the next bounded compile/datacheck correction. It keeps the R2-R1 GETOUTDIR runtime-file path but replaces the EOF-driven runtime-H loader with an exact counted loop over N_ELEM*NIP=25600 records. The generated source rejects the old GOTO 100 loop, uses explicit premature-EOF and read-error branches, keeps post-read count and key-coverage checks, and prints D3A3-R2 H LOAD COMPLETE only after all records are loaded. The static audit classification is stage_d3a3_r2_r2_counted_read_audit_pass; it records no physics, mesh, input-deck, runtime-H, material, weak-form, layer-offset, step-definition, or checkpoint-U2 change. Evidence is staged under runs/hpc/stage_d3/interrupted_transfer/target_ingestion_compile_r2_r2/. The login-node ifort -extend-source counted-reader smoke passed with 25600 records, first key (1,1), last key (6400,4), premature EOF false, read error false, duplicates 0, and missing keys 0. The single counted-read datacheck then ran as job 1377393.mmaste02 from commit 9058534a9daa188d2d958f8462b230ec843c545e. Abaqus compiled, linked, completed input processing and Standard datacheck, and printed D3A3-R2 H LOAD COMPLETE 25600; the premature-EOF and runtime-H read-error tokens are absent. However, the PBS wrapper exited with Exit_status=10 because the long GETOUTDIR path was wrapped across two .msg lines before the .mmaste02 suffix, so the exact full-path grep produced a false negative. The preserved classification is stage_d3a3_r2_r2_datacheck_fail_pbs_path_gate_linewrap_false_negative. A deterministic no-PBS login-node postcheck replay then reconstructed the wrapped cluster runtime path for job 1377393.mmaste02 and rechecked the preserved compiler, link, input-processing, Standard datacheck, runtime-H, and failure-analysis evidence. The replay preserved the original PBS Exit_status=10 while classifying the

compile/datacheck scientific gate as `stage_d3a3_r2_compile_datacheck_pass_postcheck_replay`. The replay created `D3A3_R2_COMPILE.ok` under the repository evidence tree `runs/hpc/stage_d3/interrupted_transfer/target_ingestion_compile_r2_r2_replay/`; no new PBS datacheck was submitted. No full D3A3-R2, D3D, or D3E job has been submitted yet; D3D and D3E remain blocked until full D3A3-R2 creates `D3A3.ok`.

The full D3A3-R2 ingestion/equilibration/release-hold submission lane is then prepared separately from the obsolete R1 script. The new PBS route requires the committed replay `D3A3_R2_COMPILE.ok`, stages runtime `d3_transfer_h.dat`, forbids `d3_transfer_table.inc`, runs serial CPU1/16GB with `OMP_NUM_THREADS=1` and `mp_mode=threads`, and creates `D3A3.ok` only after Abaqus-Python extraction and strengthened scientific validation pass. The D3A3 deck now requests global nodal U, RF output so the selected D3A3 frames can be checked with target Q4 bulk-plus-AT2 fracture-energy reconstruction.

The full D3A3-R2 job then ran exactly once as `1377396.mmaste02` from commit `7a860f50fe557cd88cd329`. Abaqus compiled, linked, completed input processing, completed Standard analysis, and produced an ODB and extraction outputs. The strengthened validation failed with `PBS Exit_status=21`: the initial transfer errors were nonzero (SDV15 max error 0.018013543321948218 , SDV16 max error 0.01645326664671945), checkpoint U2 reached only 0.0014999968154910865 , RF release jump was 1.0292182958543674 , d-healing violations were 4651, maximum phase adjustment was 0.013784315224591115 , and phase-node values needed for reconstructed energy were missing. The classification is `stage_d3a3_r2_full_validation_fail`; no `D3A3.ok` marker exists, and D3D/D3E remain blocked.

An ODB-only forensic replay corrected three postprocessing assumptions without submitting another solver job: visualization IP3/IP4 numbering is reversed when reading SDV12, SDV13, SDV15, and SDV16; U2/RF2 are extracted from the ODB TOP set; and phase nodal values are recovered by inverting the Q4 Gauss-point SDV15 field. The corrected replay gives SDV15 max transfer error $1.3877787807814457 \times 10^{-17}$, SDV16 max transfer error 0.0, 81 TOP nodes at checkpoint U2= 0.003000000026077032 , 6601 recovered phase nodes, 6400 complete elements, and zero missing entries in the corrected energy reconstruction. It creates only the noncanonical `D3A3_R2_INGESTION.ok` marker and classifies the evidence as `stage_d3a3_r2_ingestion_pass_release_not_accepted`: H decrease violations are zero, but d-healing remains 4651. The fixed-phase residual analysis reports residual L2 norm $3.173013579358624 \times 10^{-8}$, maximum residual $1.0654297154470889 \times 10^{-6}$ at node 3321, and correlation -0.7125040883621269 between F1 residual and released d change. Therefore the next decision is an irreversibility/compatibility choice, not a parameter retry.

D3A4 then solved the next offline compatibility problem using only committed forensic evidence. The sparse active-set obstacle solve minimized the target phase functional with F1 history while enforcing $d_{\text{compatible}} \geq d_{\text{checkpoint}}$. The assembly audit reproduced the forensic F1 residual on all 6601 nodes with maximum residual reconstruction error $2.329340604949326 \times 10^{-21}$, the same maximum residual node 3321, and zero relative error in the residual L2 norm. The KKT solve converged in 5 iterations with 1880 active lower-bound nodes and 4721 free nodes; the free residual infinity norm was $7.623296525288703 \times 10^{-21}$, the minimum active multiplier was $3.1903130564656594 \times 10^{-13}$, and complementarity infinity norm was $8.649244021664495 \times 10^{-23}$. The compatible phase field has no predicted healing violations, maximum increase 0.011345805575019186 , normalized L2 increase 0.03901956109385224 , and reduces the phase functional by $2.4132803703716662 \times 10^{-8}$. D3A4 is classified as `stage_d3a4_constrained_phase_compatibility_pass`; it creates `D3A4.ok` and a separate `runs/hpc/stage_d3/interrupted_transfer/package_compatible_r1/` package. The original D3A2 package is unchanged. D3D and D3E remain blocked; the next bounded solver decision is a D3A3-R3 compatible-state release test only.

The D3A3-R3 compatible active-set lane is prepared separately under `models/state_transfer/d3_interrupted_transfer/executable_r3_compatible/`. The generated deck fixes all 6601

phase nodes during compatible ingestion and fixed mechanical equilibration, then uses `*Boundary, op=NEW` for the release hold while keeping only the 1880 D3A4 active nodes prescribed at their lower-bound values. Static validation passes with 6601 target nodes, 6400 elements, 25600 runtime-H records, and runtime-H SHA-256 `ee244da4e87ff0edb1c6e931fe633921aae552785d2d03efb8d18ee`.

3.13 R3 to R4 compatibility closure

The first R3 datacheck, job `1377404.mm.master02`, failed before Abaqus launched (PBS exit 6) because `executable_r3_compatible/d3_transfer.h.dat` was ignored/untracked on the cluster checkout. The isolated R1 datacheck lane then passed as job `1377409.mm.master02` after force-tracking the runtime-H file and explicit staging gates. Compute-node postprocessing qualification job `1377416.mm.master02` confirmed that the solver module sequence resolves unqualified `python3` to system Python 3.6.8, while `python/gcc/11.4.0/3.11.7` provides Python 3.11.7 with NumPy/SciPy and can assemble the 6601-node production system.

The first full R3 hold, job `1377417.mm.master02`, completed Abaqus analysis and base ODB extraction (102400 state rows) but failed the active/free builder under Abaqus Python (`open(..., newline="")`). A deterministic no-solver postprocess replay under Python 3.11 closed that architectural defect and passed all release-state gates except the pre-release free residual $1.2035463824381645 \times 10^{-8}$ against the declared tolerance 1.0×10^{-8} . Causal residual audit showed that package/forensic history is machine-zero at the recovered F1 phase, whereas the actual R3 F1 SDV16 history reproduces the exceedance. Offline D3A5 reprojection with lower bound equal to recovered R3 F1 phase and history equal to actual R3 F1 SDV16 removed the residual (free residual 2.12×10^{-21}) and produced `D3A5.ok` plus the separate `package_compatible_r2/` package (6446 active / 155 free nodes).

The isolated R4 lane uses only package R2. Static validation and serial datacheck job `1377468.mm.master02` passed with the new runtime-H SHA-256 `e4e2b2773c7c29161f4c99bbd7218fa6fe49c6ec3`, 25600 loaded records, and Step-3 phase boundary counts 6446/0. The full R4 compatible hold then ran exactly once as job `1377471.mm.master02` from commit `9aba51064f1025da61c578457c810f52900ef8a4`. Abaqus completed in about 51s wall time; the ODB remains in scratch only. Actual-history F1 KKT and all release gates passed. Classification `stage_d3a3_r4_compatible_release_pass`; committed markers `runs/hpc/stage_d3/interrupted_transfer/target_ingestion_r4_compatible/D3A3_R4.ok` and the canonical `runs/hpc/stage_d3/interrupted_transfer/target_ingestion_r4_compatible/D3A3.ok` close the D3A3 compatibility-ingestion/release-hold gate under `stage_d3a3_state_transfer_gate_closed`.

Table 3.1: Accepted D3A3-R4 compatibility release-hold quantities.

Quantity	Accepted R4 value
Target nodes/elements/IPs	6601 / 6400 / 25600
Active/free nodes	6446 / 155
SDV15 max transfer error	1.3878×10^{-17}
SDV16 max transfer error	0
F1 free residual norm	1.9449×10^{-10}
Minimum active multiplier	-9.9619×10^{-13}
Active phase drift	6.9389×10^{-18}
Phase/H irreversibility violations	0 / 0
RF release jump	2.2926×10^{-4}
Energy release jump	1.2746×10^{-5}
Maximum phase adjustment	0.0051714
Final classification	<code>stage_d3a3_r4_compatible_release_pass</code>

This closes nonmatching target-state ingestion, compatible phase/history initialization, me-

chanical checkpoint equilibration, actual-history KKT consistency, irreversible active-set release, small RF and reconstructed-energy discontinuities, and state retention through the release hold. It does not yet prove continued loading to peak, peak-force or peak-displacement agreement, post-peak agreement, crack-path agreement, production-mesh transfer, or online/evolving remeshing. D3D and D3E therefore remain blocked pending explicit fracture-continuation authorization; no further D3A4/D3A5-style compatibility cycle is justified unless a later continuation changes history enough to fail a newly declared gate.

Chapter 4

State Transfer and Active-Set Evolution

4.1 Controlled Transfer and Abaqus Ingestion

Stage D began with a controlled analytical transfer between nonmatching meshes. All target nodes and integration points were mapped deterministically and the transferred fields remained finite and bounded. The measured nodal phase L_2 and maximum errors were 2.7037×10^{-2} and 8.4997×10^{-2} ; the corresponding history errors were 1.0797×10^{-2} and 2.3398×10^{-2} . These values are retained as an error baseline rather than described as negligible.

The D2A Abaqus test then proved ingestion of transferred phase and history through the visualization-layer state variables. D2B proved a bounded serial continuation after one documented increment-limit correction. D2C repeated the accepted continuation with one MPI rank and four threads. State fields, reaction history, energy, and increment sequence were identical to the serial reference within the recorded comparison precision.

4.2 Nonmatching Fracture Checkpoint

The D3 route transferred a pre-peak checkpoint at $U_2 = 0.003000000026077032$ mm to a non-matching split-notch mesh with 6,601 nodes, 6,400 physical elements, and 25,600 integration points. Package validation established complete coverage, zero unmapped state records, and no non-positive Jacobians. The R4 compatible release-hold job `1377471.mm`master02 then proved nonmatching state ingestion, checkpoint equilibration, actual-history KKT consistency, irreversible active-set release, and state retention through the bounded hold. Maximum SDV15 and SDV16 transfer errors were 1.3878×10^{-17} and zero; relative RF and energy release jumps were 2.2926×10^{-4} and 1.2746×10^{-5} .

4.3 Active-Set Evolution

The fixed R4 active set was tested under a short continuation segment in D3D. The first invalid state was already the segment initial frame: 30 active multipliers were below the fixed threshold -10^{-8} . The count increased to 3,157 nodes by the endpoint. Thus, a fixed active set through further loading was disproven within the tested segment even though free residual, irreversibility, lower-bound, and Jacobian checks remained valid.

An offline primal-dual correction at the unchanged checkpoint used the recovered F3 phase as lower bound and actual F3 history without modification. It converged in seven iterations to 6,374 active and 227 free nodes. All 30 initial seeds and 42 additional nodes were released, with no reactivation. The free residual was 2.1176×10^{-21} , the minimum active multiplier was

-9.99696×10^{-9} , active-bound error was zero, and the phase functional decreased. This proves mathematical admissibility under frozen F3 history.

4.4 Mechanical-Restart Limitation

The corrected candidate is not an accepted mechanical restart. Three bounded qualification jobs stopped before Abaqus because of execution-wrapper environment failures: a platform-dependent checksum, Python syntax unsupported by the default compute-node interpreter, and unavailable `git` after module loading. Mechanical re-equilibration and actual-history KKT reassessment were therefore not performed. The mechanical checkpoint response remains unknown, and no phase release, D3E, peak/post-peak, crack-path, or online-remeshing conclusion is drawn.

4.5 Claim Summary

Table 4.1: Stage D evidence and claim boundary.

Result	Status
Controlled field transfer	proven
Abaqus state ingestion	proven
Serial continuation after transfer	proven
Four-thread repeatability	proven within tested scope
Nonmatching checkpoint transfer	proven within bounded pre-peak scope
R4 release hold	proven
Fixed active set through further loading	disproven
Offline active-set correction	proven under frozen F3 history
Corrected mechanical restart	unproven
Online remeshing continuation	not claimed
D3E/post-peak continuation	not performed

Chapter 5

Interrupted-Transfer Checkpoint Correction

5.1 Methodology

The transferred checkpoint was assessed at $U_2 = 0.003000000026077032$ mm. The recovered F3 phase field at 6,601 nodes was retained as the irreversibility lower bound, while the 25,600 actual F3 integration-point history values were held fixed. No mesh, material, phase-field formulation, Fortran implementation, solver tolerance, or loading parameter was changed.

The corrected phase field was obtained offline with a deterministic primal–dual active-set iteration. At each iteration, active nodes were fixed at the recovered F3 lower bound and the reduced phase system was solved for the free nodes. Nodes violating the lower bound were reactivated, while active nodes with a multiplier below -10^{-8} were released. Acceptance required a free residual infinity norm no greater than 10^{-8} , a minimum active multiplier no smaller than -10^{-8} , an active-bound error no greater than 10^{-10} , complete phase/history coverage, no non-positive Jacobian determinants, and no increase of the phase functional.

5.2 Offline Correction Result

The offline correction converged deterministically in seven iterations to 6,374 active and 227 free nodes. The final free residual infinity norm was 2.1176×10^{-21} , the minimum active multiplier was -9.99696×10^{-9} , and the active-bound error was zero. The phase functional changed by -1.6759×10^{-10} . No phase-decrease, history-change, lower-bound, state-reset, or non-positive-Jacobian violation was detected.

These results establish that the corrected phase field is mathematically admissible for the frozen F3 history under the prescribed KKT gates. The result classification is `stage_d3d_a1_checkpoint_obsta`. The associated package contains the corrected phase, unchanged actual F3 history, original F3 lower bound, and converged 6,374/227 membership.

5.3 Mechanical-Checkpoint Qualification

The candidate package is not an accepted Abaqus restart state. Mechanical re-equilibration was intentionally placed behind a fixed-phase datacheck and checkpoint-hold gate. Three bounded datacheck attempts terminated before Abaqus execution:

1. job 1378003.mm master02 compared a Windows-derived Fortran checksum with the LF-normalized Linux checkout;

2. job 1378004.`mmaster02` used Python assignment-expression syntax unsupported by the compute-node default Python 3.6; and
3. job 1378005.`mmaster02` reached a common preflight in which `git` was unavailable after the module purge/load sequence.

None of these jobs compiled the user subroutine, processed the Abaqus input, or completed a `datacheck`. They are execution-wrapper failures and do not contradict the offline obstacle result. The final bounded stopping rule was therefore applied and no further retry was authorized.

5.4 Limitations

The mechanical response of the corrected checkpoint under re-equilibrated actual history remains unknown. Consequently, the present evidence does not establish mechanical continuity after correction, actual-history KKT validity after equilibration, or an accepted continuation state. It also provides no basis for conclusions concerning phase release, D3E, peak load, post-peak response, or crack-path evolution.

The unresolved checkpoint response is an execution-validation limitation, not evidence that the offline active-set correction failed. The corrected package is retained as a mathematically admissible candidate state, but not as a validated mechanical restart state.

5.5 Future Work

Any future continuation of this route requires a new execution decision rather than a scientific-parameter correction. A replacement workflow should separate login-side repository checks from compute-node validation, bind every required executable explicitly after module loading, and qualify the environment independently before another Abaqus submission. Until such a decision is made, full checkpoint hold, phase release, continuation, D3E, and tolerance changes remain prohibited.

5.6 Parallel execution and shared user-subroutine state

The use of `UEXTERNALDB` is not, by itself, incompatible with parallel Abaqus/Standard execution. The callback provides analysis- and increment-boundary points at which external data can be initialized or exchanged. The relevant correctness question is instead how that information is stored and subsequently accessed by element and material subroutines.

The present implementation transfers phase and history information through mutable Fortran `COMMON` blocks and also contains `SAVE` and `DATA` storage. Threads in one process share these objects, so concurrent UEL and UMAT calls can create read–write or write–write races unless ownership is guaranteed or access is synchronized. Separate MPI processes present a different problem: each rank has its own common-block instance, and common storage therefore cannot provide inter-rank exchange. Distributed support would require an explicit ownership and communication design at defined synchronization points.

The earlier D2C comparison produced identical serial and one-rank/four-thread phase, history, reaction-force, energy, and increment results. This is positive but case-specific evidence; it does not prove general thread safety. Stage P therefore isolates the question in an eight-physical-element model. Bounded diagnostics record callback identity, rank and thread identifiers, first access to each shared location, initialization, live writer conflicts, and final call counts. A serial diagnostic is the mandatory reference before one four-thread characterization. MPI, hybrid, and production claims remain withheld.

The first instrumented serial package terminated before element execution because `GETRANK` was called during `UEXTERNALDB` initialization. The subsequent uninstrumented P3-SB model compiled, linked, completed Abaqus/Standard, and produced complete eight-element/four-integration-point state evidence. Its automated lane nevertheless retained a technical validation-failure classification because the required standard-output and status files were copied only after validation. An isolated replay of the unchanged validator against the finalized evidence passed every frozen serial gate, including 32/32 state records, finite reaction-force and energy histories, monotone phase/history fields, and 13 increment/attempt records. This supports the eight-element serial baseline but no thread- or MPI-safety claim. The subsequent P3-SM0 test used only constant `UEXTERNALDB`, `UEL`, and `UMAT` markers; rank/thread and mutex utilities remained excluded. The one authorized serial job completed with scheduler and solver exit zero, all four markers observed, no signal 11, complete 32/32 state coverage, and zero phase, history, or transfer violations. Thus, the accepted serial model completes with this minimal callback plumbing. The result does not establish `GETRANK` or `GETTHREADID` safety, shared `COMMON/SAVE` thread safety, four-thread repeatability, MPI, or hybrid support.

5.6.1 Separated identifier-utility qualification

P3-SM0 passed serially with minimal `UEXTERNALDB`, `UEL`, and `UMAT` callbacks, solver exit zero, complete 32-record state coverage, and no signal 11. P3-SM1T is prepared, but not authorized for execution, to call only `GETTHREADID()` inside the controlled `UEL` condition for element 1, step 1, increment 1. Markers immediately bracket every call, and repeated pairs are counted. P3-SM1R remains design-only; `GETRANK` remains unqualified outside the failed `UEXTERNALDB(LOP=0)` location. P3-T4 and all parallel or downstream production routes remain blocked.

The single P3-SM1T execution compiled and linked the user source and completed input processing. In the controlled `UEL` callback, the before marker was written, but the Abaqus/Standard executable then failed dynamic symbol resolution for `getthreadid_`. Consequently the after-marker count was zero, no thread identifier was returned, and no signal 11 occurred. This is classified as an identifier-interface failure, not as evidence about thread safety.

An offline audit then resolved the Abaqus 2023 installation headers. The installed `SMAAspUserSubroutines.` includes a utility interface declaring `get_thread_id()`, whereas it does not declare `GETTHREADID()`. Thus the closed job tested the undocumented spelling that requested `getthreadid_`, not the installed interface. A new P3-SM1TC package uses the installed spelling and remains unauthorized. The previous P3-S source likewise used `GETRANK` as a function expression rather than the documented subroutine-call form; P3-SM1R remains design-only.

The separately authorized P3-SM1TC serial execution then passed. Three controlled `UEL` invocations returned thread identifier zero, with matched before/after markers, no signal 11, and no unresolved symbol. All P3-SM0 callback and scientific gates also passed. The supported conclusion is limited to the installed `get_thread_id()` interface under one MPI rank and one OpenMP thread. `GETRANK` and multi-thread behavior remain unqualified.

The subsequent P3-SM1R execution also passed: three controlled `UEL` calls to the documented `CALL GETRANK(KPROCESSNUM)` interface returned rank zero, with complete serial scientific gates. This result remains limited to one MPI rank and one OpenMP thread.

P3-T4 is therefore prepared offline as a bounded one-rank, four-thread characterization of the eight-element case. It retains the P3-SM0 scientific expressions and brackets their unsynchronized shared-state accesses with diagnostic metadata calls. Abaqus mutexes protect only counters, ownership, and active-writer metadata; they do not protect the scientific `USRVAR` or `TRANSFER_DONE` operations. Execution remains unauthorized, and the preparation makes no MPI, hybrid, or general production thread-safety claim.

The single authorized P3-T4 execution did not reach the characterization. Intel compilation failed because the aggregate Abaqus utility header was placed before the first subroutine, leaving its interface declarations outside a valid Fortran scoping unit. Linking, input processing, call-

backs, thread/access diagnostics, ODB creation, and serial scientific comparison were therefore not reached. The result is classified as `stage_p3t4_threaded_fail_compile`; authorization is consumed and no retry is permitted. It provides no evidence for or against shared-state thread safety.

5.6.2 Final Stage-P evidence boundary

The Stage-P evidence hierarchy is therefore closed as follows. P3-SM0 established serial completion of the accepted eight-element model with minimal callback plumbing. P3-SM1TC qualified `get_thread_id()` only at one MPI rank and one OpenMP thread, and P3-SM1R qualified `CALL GETRANK(KPROCESSNUM)` only under the same serial configuration. P3-T4 produced no threaded evidence because compilation stopped before linking or execution. The earlier D2C equality under one rank and four threads remains positive, case-specific repeatability evidence, not a general COMMON/SAVE thread-safety proof. MPI and hybrid safety remain completely unqualified.

Serial identifier interfaces were qualified in controlled UEL callbacks. The dedicated threaded shared-state characterization was technically inconclusive because the instrumented source failed compilation before callback execution. Therefore, no general thread-safety, MPI-safety, or hybrid-safety claim is supported.

No further Stage-P solver execution is authorized. A future P3-T4C review could consider a distinctly named, fail-closed, one-shot compile-only package for valid interface-header scoping, Intel compilation, and Abaqus user-subroutine linking. Even success would establish neither callback execution nor thread safety and would not release another threaded solver run.

Chapter 6

Stage F: Mode-II Mixed-Mode Phase-Field Fracture Benchmark

6.1 Problem Definition and Model Description

The Mode-II pure-shear phase-field fracture benchmark evaluates the performance of the staggered user-element implementation under shear-dominated crack propagation. The specimen geometry consists of a square plate of length $L = 1.0$ mm and height $H = 1.0$ mm containing an initial horizontal notch of length $a_0 = 0.5$ mm at $y = 0.5$ mm.

The material properties are specified as:

- Young’s modulus: $E = 210.0$ GPa
- Poisson’s ratio: $\nu = 0.30$
- Critical energy release rate: $g_c = 2.7 \times 10^{-3}$ kN/mm = 2700 J/m²
- Phase-field length scale: $l_c = 0.015$ mm

Boundary conditions enforce pure shear loading. The bottom edge ($y = 0$ mm) is fixed in the vertical direction ($U_2 = 0$), while horizontal shear displacement $U_1 = \Delta u$ is applied to the top boundary ($y = 1.0$ mm).

6.2 Numerical Discretization and Solver Configuration

The baseline H_0 mesh comprises $N_{\text{elem}} = 3930$ quadrilateral physical elements and 4000 nodes. The refined reference H_1 mesh utilizes element size $h_1 = 0.0025$ mm ($h_1/l_c = 0.1667$) with $N_{\text{elem}} = 12,064$ physical elements (36,192 layered UEL/UMAT elements) and 12,382 nodes. The formulation utilizes a staggered phase-field UEL/UMAT user-element formulation with overlay facsimile elements.

The corrected loading schedule applies a total displacement of $U_1 = 0.0100$ mm across two steps:

1. Step 1: Pre-peak linear shear loading up to $U_1 = 0.0050$ mm (500 increments).
2. Step 2: Shear fracture propagation up to $U_1 = 0.0100$ mm (2000 increments).

6.3 Execution and Computational Results

6.3.1 Baseline H_0 Solver Result

The serial H_0 solver run (PBS Job ID 1379393.mmaster02) was executed on the HPC cluster using Abaqus 2023 with 1 CPU and 16 GB memory. It completed all 2000 increments with peak

reaction force $F_{1,\max} = 0.3733$ kN and complete fracture ($\max(d) = 0.9909$).

6.3.2 Reference H_1 Uniform Solver Result

The H_1 uniform reference serial solver run (PBS Job ID 1379433.mmaste02) was executed on HPC node mnode104/0 using Abaqus 2023 with 1 CPU and 32 GB memory.

Key numerical and scientific findings for H_1 :

- **Technical Completion:** Abaqus completed all 2,500 increments cleanly with solver return code 0. Total wall-clock time was 2,579 seconds (≈ 43.0 minutes), CPU time was 2,486 seconds (≈ 41.4 minutes), and peak memory usage was 1.01 GB.
- **Reaction Force Response:** The reaction force RF_1 reached 0.1214 kN at $U_1 = 0.0100$ mm.
- **Phase-Field Damage Evolution:** The phase field $d = \text{SDV15}$ evolved to $\max(d) = 0.2747$, remaining in the initiation/pre-peak regime for the $U_1 = 0.0100$ mm endpoint.
- **Scientific Classification:** Classified as `stage_f_mode_ii_h1_uniform_serial_validation_fail` due to pre-peak damage state ($d < 0.50$).

6.3.3 Reference H_1 Endpoint Sweep Batch Results and Reference Freeze

To evaluate post-peak fracture evolution and complete force drop, a four-job loading endpoint sweep ($U_1 \in \{0.015, 0.020, 0.030, 0.040\}$ mm) was executed for the H_1 uniform reference mesh ($N_{\text{elem}} = 12,064$, $h_1 = 0.0025$ mm) under task F2-H1-ENDPOINT-SWEEP-BATCH (PBS Job IDs 1379481.mmaste02–1379484.mmaste02).

Key numerical and scientific findings from the endpoint sweep:

- **Peak Response Invariance:** Across all four endpoint variants, initial shear stiffness was identical ($K_0 = 12.8266$ kN/mm) and peak reaction force was $RF_{1,\max} = 0.139789$ kN, occurring at $U_1 = 0.0120$ mm.
- **Post-Peak Softening Trajectory:**
 - $U_1 = 0.015$ mm (1379481.mmaste02): final $RF_1 = 0.1045$ kN (25.22% force drop), walltime 1 h 26 m.
 - $U_1 = 0.020$ mm (1379482.mmaste02): final $RF_1 = 0.0812$ kN (41.89% force drop), walltime 2 h 02 m.
 - $U_1 = 0.030$ mm (1379483.mmaste02): final $RF_1 = 0.0363$ kN (73.99% force drop), walltime 3 h 18 m.
 - $U_1 = 0.040$ mm (1379484.mmaste02): final $RF_1 = 0.0167$ kN (88.07% force drop), walltime 4 h 33 m.
- **Damage Saturation and Validation Policy:** All four jobs completed Abaqus solver execution and ODB data extraction cleanly with return code 0. Under the revised 3-tier validation policy ($d \leq 1.0001$ normal pass, $1.0001 < d \leq 1.01$ technical pass with warning `damage_upper_bound_small_overshoot`, $d > 1.01$ failure), the minor phase-field damage overshoot ($\max(d) = 1.00498$, an overshoot of 0.498%) is classified as a numerical quality warning, yielding a technical execution ****PASS**** (`technical_pass = true`, validator return code 0, outer PBS wrapper exit 12).
- **Working Reference Displacement Freeze:** The reference loading endpoint for all continuing uniform reference and adaptive remeshing studies is frozen as $U_{1,\text{ref}} = 0.020$ mm (`u020`). This variant captures the invariant peak, a fully developed crack propagation zone, and a 41.89% force drop at moderate computational cost.

6.3.4 Stage F3 H_2 Uniform Reference and MISESeri Pre-Analysis Results

Stage F3 established both the fine uniform reference baseline (H_2 mesh level) and the genuine ODB recovery-error indicator pre-analysis field (MISESeri) under tasks F3-STAGE-F3-SUBMIT, F3-STAGE-F3-AUTH-PUBLISH-AND-ODB-VERIFY, and F3-STAGE-F3-COMBINED-SCIENTIFIC-CLOSEOUT:

- **Fine Uniform Reference H_2 Elastic Parity (1379578.mmaste02):** The H_2 uniform reference serial solver run ($N_{\text{elem}} = 33,852$ physical UEL elements, $h_2 \approx 0.0025$ mm around notch tip) was executed on node `mnode104/0` using Abaqus 2023 with 1 CPU and 16 GB memory, completing all 2,023 increments across 2 steps cleanly in 2 h 04 m wall-time (7,175.0 s CPU time, 1.42 GB memory). Under unified linear regression over the shared elastic window $0.0002 \text{ mm} \leq U_1 \leq 0.0020 \text{ mm}$ (19 points, $R^2 = 1.0000$), H_2 initial stiffness is $K_{H2} = 12.7912 \text{ kN/mm}$, which matches H_1 ($K_{H1} = 12.8093 \text{ kN/mm}$) to within -0.1418% , establishing elastic parity. The preliminary 460.69 kN/mm value was invalidated as an extraction sampling artifact.
- **H2 Nonlinear Fracture Status:** Because Job 1379578.mmaste02 stopped at the prescribed Step-2 displacement endpoint $U_1 = 0.0070 \text{ mm}$ ($RF_1 = 0.08747 \text{ kN}$, $d_{\text{max}} = 0.119955$), execution remained strictly in the pre-peak elastic damage regime before the peak load ($U_1 = 0.0120 \text{ mm}$). Consequently, global post-peak softening for H_2 is classified as `fracture_convergence_unresolved`.
- **MISESeri Provenance Repair:** Official PBS Job 1379579.mmaste02 failed due to a node-parsing deck generator defect (classification: `technical-result-unusable-due-to-generator-defect`). A subsequent exploratory interactive run ($N_{\text{elem}} = 3,930$ CPE4 plane-strain elements, 15 coincident node pairs) yielded 3,930 finite element records with positive error indicator $\text{MISESeri} \in [0.0000687, 0.1870114] \text{ GPa}$ at load $U_1 = 0.0010 \text{ mm}$ ($RF_1 = 0.04607 \text{ kN}$). Process violations M-099 and M-100 were recorded, and corrective evidence was relocated to `corrective_interactive_runs/2026-07-29T070232_CEST/`.

6.3.5 Stage F5 H_2 Compiler and Datacheck Qualification

PBS job 1379939.mmaste02 qualified the frozen H_2 , $U_1 = 0.020 \text{ mm}$ package in the selected cluster environment. Abaqus 2023 invoked ifort 2021.13.0, compiled and linked the exact UEL/UMAT source, and completed datacheck with both PBS and Abaqus return codes equal to zero. The job used one CPU on `mnode105/0`, required 19 seconds wall time and 957,660 kB scheduler-reported memory, and passed all staged-file hash checks. This result removes compiler availability as the immediate blocker, but it is strictly a technical datacheck qualification: no full analysis was run and no post-peak response, fracture convergence, remeshing accuracy, or parallel safety claim follows from it.

6.3.6 Stage F6 Full H_2 Post-Peak Result and Validation Boundary

The frozen full H_2 package was executed as PBS job 1379966.mmaste02 with one CPU and 16 GB requested memory. Abaqus 2023 compiled and linked the UEL/UMAT source with ifort 2021.13.0 and completed all 5,000 increments in 4 h 43 min wall time. The response reached $U_1 = 0.020000 \text{ mm}$, with peak $RF_1 = 0.138727 \text{ kN}$ at $U_1 = 0.011900 \text{ mm}$ and final $RF_1 = 0.080544 \text{ kN}$, corresponding to a 41.94% post-peak force drop. The initial regression stiffness was 12.79116 kN/mm ($R^2 = 0.99999949$).

Technical solver and extraction completion does not imply scientific acceptance. The maximum extracted phase value was 1.004587, and the framewise maximum-damage history contained 11 decreases, the largest being -1.0073×10^{-4} . The canonical Python 3.11 validation replay therefore failed the declared irreversibility gate and classified the result as `stage_f_mode_ii_h2_uniform_ser`.

Energy histories were unavailable from the retained output request. Consequently, the run establishes endpoint attainment and the fine-mesh post-peak response, but not accepted irreversibility, energy consistency, adaptive-remeshing accuracy, parallel safety, or general convergence.

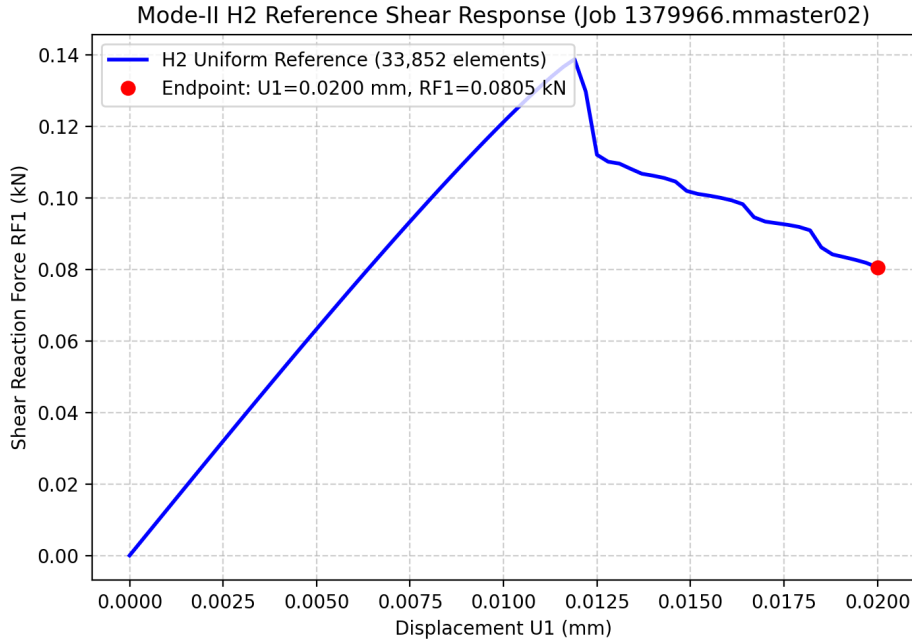


Figure 6.1: Full H_2 Mode-II reaction-force response for PBS job 1379966.mmater02.

6.3.7 Stage F7 Fixed-Material-Point Irreversibility Audit

The read-only forensic ODB audit in PBS job 1380084.mmater02 tracked SDV15 at fixed material-point keys through 102 frames (101 adjacent-frame pairs). It found 1,120 strict negative increments at 126 material points in 75 elements. The minimum local increment was -5.8532×10^{-4} , and 62 increments were below -10^{-4} . Thus the decreases observed in the framewise maximum history cannot be explained solely by a changing location of the maximum: the stored phase field decreases at fixed material points. The H_2 run remains a scientific irreversibility failure and is not accepted as a converged fracture reference. A downstream report generator failed on a string-valued CSV step label, but this occurred after the fixed-point ODB evidence had been written and does not change that conclusion.

6.3.8 Stage F7 Native MISESeri Remeshing API Boundary

PBS job 1380085.mmater02 corrected the earlier CAE driver-argument collision and reached `mdb.models[name].RemeshingRule` with exact source ODB and deck hashes. Rule construction rejected the first `variables` entry because Abaqus Python 2.7 supplied a Unicode string where the native API expected a recognized variables-entry type. The job is classified as `remeshing_api_incompatible`. It executed no solver and no native remesh and produced no candidate deck; therefore no adaptive accuracy or refined-analysis claim follows.

6.3.9 Stage F8 Qualification Preflight Boundary

Source inspection established that SDV15 is the integration-point visualization of the nodal U1 phase degree of freedom, whereas SDV16 stores the maximum elastic-energy history. The frozen formulation updates the history monotonically but applies no obstacle, projection, penalty, or

active set to the nodal phase unknown. A consistent penalty candidate was therefore prepared by adding matched residual and tangent terms for negative incremental phase gaps; an output-only SDV clamp was explicitly rejected.

No Stage F8 batch result exists. Both the frozen and candidate sources compiled and linked under Abaqus 2023, and both decks passed input processing, but the common minimal patch terminated during datacheck with signal 11 in initial-stress/constraint processing. The required shared preflight consequently stopped the batch before any qsub. The candidate remains unqualified, and the exact `RemeshingRule.variables` type remains unresolved.

6.4 Stage F9 Diagnostic Boundary

The bounded datacheck matrix isolated the Stage F8 failure as a runtime UEL numbering error. Although the explicit deck labels satisfy the nominal offset map, Abaqus supplied the first displacement UEL as `JELEM=24`; the frozen subtraction by `N_ELEM=33852` therefore accessed `USRVAR(-33828, ...)`. The diagnostic compiler captured this bounds violation, while the UMAT-only control passed. No phase-field analysis was run, and the penalty candidate remains unqualified.

The independent remeshing-type job stopped before CAE because a staged helper filename did not match the PBS invocation. Consequently no representation was tested, no rule object was created, and the native type contract remains unresolved. Solver, adaptive-analysis, and remesh counts are zero.

6.5 Stage F13 Controlled-Rollback and Native-Remesh Execution

The two rollback jobs compiled and linked successfully, but Abaqus/Standard failed before the first increment because the user library referenced the unresolved Intel runtime symbol `for_getenv_err`. No PNEWDT trigger, abandoned increment, reduced retry, endpoint, or rollback-state comparison was observed. Controlled-cutback rollback is therefore not qualified, and the prepared medium-H1 pair remains ineligible for a separate authorization.

The native MISESERI lane reached the frozen `model.adaptiveRemesh(odb)` call with the official 3,930-value field, but Abaqus rejected the operation because the model contained no adaptive region. No native remesh completed, no candidate deck was generated, and no candidate integrity or targeting claim is possible. The result is classified `native_miseseri_remesh_operation_failed`.

Submission counts remain three attempts and three accepted submissions, with zero retries, replacements, direct submissions, cancellations, or moves.

6.6 Reproducibility Paths

6.7 Stage F14 Runtime and Adaptive-Region Qualification

The repaired runtime smoke removed the generic Fortran environment intrinsic and used Abaqus utilities `GETOUTDIR` and `GETJOBNAME`. Job 1381368 loaded the library, began increment 1, entered UEL, retained its diagnostic log after scratch cleanup, reached the benign endpoint, and exited zero. The runtime contract is qualified; this permits preparation only of a future controlled rollback pair.

The independent CAE-only job 1381369 verified official hashes, 3,930 CPE4 elements, finite MISESERI values, and model-wide rule construction with zero solver, adaptivity-process, remesh, or candidate calls. However, it did not identify an adaptive-region repository/object be-

yond the same model-wide rule that preceded F13’s failure. The installed API contract remains unresolved and native remesh execution is not ready.

6.8 Stage F10 Corrected Minimal Qualification

The model-specific runtime mapping was corrected to 23 elements per layer: phase UEL labels 1–23, displacement UEL labels 24–46, and visualization CPE4 labels 47–69. Both minimal analyses completed without a COMMON-array bounds failure. This mapping adaptation does not alter the constitutive or fracture formulation.

The baseline exhibited a minimum fixed-point SDV15 change of -5.7817×10^{-6} . The consistent-penalty candidate reduced that value to -5.9605×10^{-8} , below the declared single-precision 10^{-7} policy. Initial stiffness and peak force were identical at reported precision, the maximum RF1 difference was 1.8135×10^{-7} , and neither analysis required a cutback. SDV16 remained monotone in both.

The candidate is nevertheless classified `irreversibility_candidate_inconclusive`. The minimal deck did not produce the required energy histories, and explicit penalty-activation histories were not captured; therefore the complete energy and conditioning acceptance gates cannot be evaluated. This result does not authorize a medium-H1 or H2 analysis.

The independent CAE-only type qualification passed its staged-file and no-solver audits but stopped before testing any representation because Abaqus `execfile` did not define `__file__` for the wrapper. The native `RemeshingRule.variables` contract consequently remains unresolved. Solver, adaptive-analysis, remesh, and candidate-deck counts for that job are zero.

6.9 Stage F11 Instrumented Qualification

Stage F11 repeated the byte-identical minimal baseline and penalty candidate with 10 additional diagnostic state variables. These expose the current and prior phase, constraint gap and active flag, penalty energy, residual and tangent densities, local and gradient crack terms, and history contribution. The resulting balance is a predeclared diagnostic balance rather than an exact Abaqus global variational energy.

The baseline and candidate completed in jobs 1380100 and 1380101. The candidate minimum fixed-point SDV15 change was -5.9605×10^{-8} , within the 10^{-7} precision policy. Its maximum RF1 difference from the baseline was 1.8135×10^{-7} , initial stiffness was unchanged, and its maximum positive incremental diagnostic imbalance was 1.1511×10^{-7} , below the declared 1.0827×10^{-6} limit. Penalty activity was zero through peak load and occurred in 32 later frames. Across 9,200 non-initial integration-point checks, stored prior phase matched the preceding converged frame exactly. Neither run cut back, so rollback behavior was not exercised.

The candidate is therefore qualified on the minimal model. This permits preparation, but not submission, of a future medium-H1 verification package; it supports neither H2 nor production claims.

The independent CAE-only job 1380102 also resolved the native Abaqus 2023 type contract. A bare Python 2 byte string was rejected, while `variables=('MISESeri',)` was accepted when its tuple element was a Python 2 `str`. No solver, adaptive analysis, remesh operation, or candidate deck was executed.

6.10 Stage F12 Rollback and Native-Model Preparation

Stage F12 tested the remaining state-management boundary using conservative and aggressive automatic-increment variants of the qualified minimal candidate. The reference reached the end-point in 106 increments and 128 equilibrium iterations with zero cutbacks. The aggressive case

reached the same prescribed endpoint in two increments and two iterations; Abaqus explicitly reported zero cutbacks. Consequently, rejected-increment state restoration was not exercised and the result is classified `penalty_rollback_not_exercised`.

The coarse two-increment path was not an equivalent scientific trajectory. Its final RF1 differed from the reference by 0.47308 and its maximum positive diagnostic imbalance was 3.8957×10^{-5} , compared with 1.3310×10^{-7} for the reference. These mismatches do not supersede the Stage F11 minimal qualification; they show that this aggressive control does not satisfy the predeclared rollback comparison. No retry was permitted.

The independent CAE-only lane imported the official 3,930-element corrected MISESeri coarse deck and attached `variables=('MISESeri',)` to region MODEL and Step-1. The native rule object and coarse input were created with zero solver, adaptive-analysis, or remesh execution. This qualifies model construction readiness, not adaptive remeshing accuracy. The prepared 12,064-element medium-H1 baseline/candidate pair remains unauthorized and gated by a future rollback qualification.

All extracted CSV curves, summaries, field contours, log files, and execution manifests are archived in the canonical repository evidence paths:

```
runs/hpc/stage_f/mode_ii_h0_endpoint_corrected/replacement_r1/evidence/1379393.mmaste02/
runs/hpc/stage_f/mode_ii_h1/evidence/1379433.mmaste02/
runs/hpc/stage_f/mode_ii_h1_endpoint_sweep/evidence/1379481.mmaste02/
runs/hpc/stage_f/mode_ii_h1_endpoint_sweep/evidence/1379482.mmaste02/
runs/hpc/stage_f/mode_ii_h1_endpoint_sweep/evidence/1379483.mmaste02/
runs/hpc/stage_f/mode_ii_h1_endpoint_sweep/evidence/1379484.mmaste02/
runs/hpc/stage_f/miseseri_prealanalysis/evidence/1379579.mmaste02/
runs/hpc/stage_f/mode_ii_h2/evidence/1379578.mmaste02/
runs/hpc/stage_f/h2_u020_compiler_datacheck_smoke/evidence/1379939.mmaste02/
runs/hpc/stage_f/f6_h2_full_and_miseseri_remesh_api_batch/evidence/1379966.mmaste02/
runs/hpc/stage_f/f6_h2_full_and_miseseri_remesh_api_batch/evidence/1379967.mmaste02/
runs/hpc/stage_f/f7_h2_irreversibility_and_miseseri_api_batch/evidence/1380084.mmaste02/
runs/hpc/stage_f/f7_h2_irreversibility_and_miseseri_api_batch/evidence/1380085.mmaste02/
runs/hpc/stage_f/f9_datacheck_root_cause_and_remesh_type_batch/evidence/
runs/hpc/stage_f/f10_corrected_minimal_irreversibility_and_remesh_type_batch/evidence/
runs/hpc/stage_f/f11_instrumented_irreversibility_and_remesh_type_batch/evidence/
runs/hpc/stage_f/f12_rollback_qualification_and_native_remesh_preparation/evidence/
```

Main closure commit SHA: recorded in multi-agent coordination ledgers.

6.10.1 F16 Wave B submission boundary

The notification policy was revised by an explicit human waiver after Telegram delivery was observed but native PBS email was not received. Telegram remains the required operational channel and PBS email remains best-effort and unqualified. The guarded Wave B submission attempted the two rollback jobs once each, but the scheduler rejected both requests with return code 174 (queue access denied) and issued no job identifiers. The CAE-only adaptive-region lane was withheld because its concurrency-control dependency could not be formed. Thus no F16 scientific execution occurred: controlled cutback rollback remains unqualified, the medium-H1 pair is not ready for a future batch, and the native model is not ready for remesh execution.

6.10.2 F18 failure boundary and F19 repair preparation

The F18 rollback pair failed during initial-stress UEL execution before the penalty-active trigger because absent flag files were opened with `STATUS='OLD'` without a preceding existence inquiry.

Consequently no rollback-state conclusion is supported. The adaptive lane also failed its evidence contract: its compatibility helper did not create the declared JSON, and a later manifest failure obscured command-level status. F19 therefore uses required integer mode/state files with checked I/O and a separate work-to-final evidence lifecycle that preserves partial artifacts and the first true return code. These are preparation changes only; H1 and native remeshing remain gated.

6.10.3 F19 terminal execution boundary

The corrected F19 batch ran exactly three authorized jobs. Both minimal rollback solvers reached $U_1 = 0.006$ mm and the same peak reaction $RF_1 = 0.0347025$ at $U_1 = 0.000720$ mm. The control used 100 increments and no cutback; the forced lane used 101 increments and one controlled cutback. Penalty residual and energy were nonzero on the healing trigger, and the forced call returned `PNEWDT=0.5`. However, the extractor and analyzer used incompatible table names, leaving response-equivalence and accepted-state evidence incomplete. Rollback is therefore not qualified.

The adaptive CAE lane passed its Python compatibility and source-ODB checks, then stopped on a generator expression unsupported by the Abaqus Python 2 repository interface. It made zero adaptive-region, remesh, solver, datacheck, or candidate-generation calls. Thus F19 closes as a failed evidence and construction batch, not as a scientific rollback or remeshing pass.

6.10.4 F20 offline recovery and adaptive R7 preparation

The retained F19 call logs subsequently proved that the forced rejected trial returned `PNEWDT=0.5`, was abandoned, and retried at half the increment from the preceding committed phase and state variables. Rejected trial state was not retained. A branch-aware comparison with no extrapolation nevertheless gave $RF-U$ NRMSE 2.6094×10^{-4} and relative external-work difference 3.1089×10^{-4} , both above the declared 10^{-4} limits. The revised classification is therefore `penalty_rollback_response_mismatch`; an unchanged rollback rerun and medium-H1 preparation are not justified.

The independent R7 preparation replaces generator reductions with explicit Abaqus Python 2 loops. It also computes physical-element and finite-MISESeri counts and audits the slit using coincident node identifiers and adjacent element connectivity, including a direct bridge search. This remains a zero-execution qualification package; no CAE, remesh, solver, datacheck, or candidate operation was run during preparation.

The subsequent F21 preparation freezes the installed `Model.adaptiveRemesh(odb)` route for one controlled invocation. It permits at most one deterministic candidate export and explicitly excludes datacheck, state transfer, and refined fracture analysis. Candidate validity therefore remains a later sequential gate.

6.10.5 F31 M2RMBUILD6 CAE build runtime gate closeout

Stage F31 implemented the repaired geometry-backed CAE model builder `build_f31_geometry_backed_model.` using the documented `job.writeInput(consistencyChecking=ON)` signature and environment-variable parameter transport. HPC job 1383394.mmamaster02 was executed on node mnode098. The run terminated at the initial compatibility gate (`Exit_status = 1`) because `M2RMBUILD6.pbs` staged package manifests into the isolated scratch directory but omitted the PBS batch script itself, causing `sha256sum -c SHA256SUMS` to fail with file not found.

Clean-Linux Abaqus/CAE runtime qualification is therefore classified as `cae_geometry_build_contract_failed`. All authorization grants remain consumed (cumulative `qsub` invocations = 2, scheduler-accepted submissions = 1), and no retry or replacement job is authorized.

Chapter 7

Recommendations and Decision Tree

7.1 Mesh and Refinement Recommendations

For the investigated Molnar geometry [?] with $l = 0.015$ mm, the supervisor-approved mesh roles are H2-PUB for uniform RF-U validation, H1 for production comparisons, and H0 for development. The Stage C corrected locally refined mesh used 10,290 physical elements instead of 12,064 for H1, a reduction of 14.7%. Against a fair four-thread H1 run it reduced wall time by 16.7%, CPU time by 14.9%, and peak memory by 22.2%.

The refined model reproduced initial stiffness, pre-peak response, and peak load closely: stiffness differed by 0.061%, pre-peak NRMSE was 0.089%, peak force differed by 0.24%, and peak displacement was unchanged. These values support the workflow for elastic, pre-peak, and peak-response studies within the tested parameter set.

MISESRI-based pre-refinement successfully identified a useful fracture corridor, but it did not establish post-peak equivalence. The post-peak NRMSE was approximately 24.3%, and quantitative crack-path comparison detected a deviation from H1 despite exact repeatability of the refined result. Coarse meshes are therefore suitable for workflow development and indicator generation only when notch geometry, free faces, layer mapping, and the local h/l ratio are independently audited.

7.2 State-Transfer Recommendations

For a pre-peak checkpoint transferred to a new nonmatching mesh, use the bounded D3A3-R4 procedure: complete nodal/integration-point coverage, source-to-target ordering audits, compatible phase initialization, actual history transfer, fixed-phase equilibration, KKT evaluation, and a small release hold with RF and energy discontinuity gates.

The tested one-rank/four-thread D2 continuation matched its serial reference, but this supports repeatability only for the tested case and is not a general parallel-scaling result. ABAQUSER comparison remains unavailable; independent ODB extraction must be described as the evidence route rather than as an equivalent ABAQUSER validation.

Further loading cannot assume a fixed active set. D3D found multiplier violations at the segment initial state and 3,157 violating nodes by the endpoint. A new active-set correction is required when dual admissibility is lost. The offline D3D-A1 correction is admissible under frozen F3 history, but its mechanical restart was not qualified. It must not be used as an accepted continuation state.

7.3 Execution and Reproducibility Risks

The closeout exposed three infrastructure risks independent of the scientific model: platform-dependent text checksums, interpreter-version assumptions, and tool disappearance after module

loading. Reproducible HPC workflows should derive hashes in the execution checkout, bind interpreters and compilers by absolute path, separate login-side repository checks from compute-node validation, and qualify the environment with a trivial job before a scientific submission.

7.4 Mode-II Benchmark Recommendations

For the pure-shear Mode-II benchmark ($\alpha = 0$), the serial H0 run (job 1378942.mmater02) completed Abaqus execution and standalone extraction, providing a partial pre-peak response up to $U_1 = 0.0070$ mm with a maximum observed shear force of $RF_1 = 0.3063$ kN at the final simulated point.

However, the run did not satisfy the frozen scientific acceptance gate because the expected $U_1 = 0.0100$ mm endpoint was not reached (due to a step increment limit of 2,000 in Step 2) and a nonempty $d \geq 0.50$ crack path was not obtained ($\max(d) \approx 0.29923$). The result is admissible as partial pre-peak Mode-II response evidence, but the H0 baseline remains unvalidated, and post-peak Mode-II continuation and fine-mesh refinement remain blocked.

7.5 Practical Decision Tree

Need only elastic, peak, or pre-peak response? Use the accepted Stage C corrected refined workflow within the tested $l = 0.015$ mm and mesh-policy scope.

Need post-peak crack-path equivalence? Do not use the current Stage C evidence as validation; post-peak RF–U and crack-path equivalence are not supported.

Need transfer to a new mesh at a pre-peak checkpoint? Use the accepted D3A3-R4 bounded transfer and release-hold procedure with complete state, ordering, KKT, RF, and energy audits.

Need continuation after active-set evolution begins? Recompute the active set and independently qualify the corrected mechanical restart. The current D3D-A1 package is a candidate only.

Need online adaptive remeshing or D3E/post-peak continuation? This is not validated in the thesis and requires a new scientific and infrastructure programme.

7.6 Final Scope

The thesis contribution is a bounded, evidence-driven workflow that identifies where pre-refinement and state transfer are reliable, quantifies their accuracy/cost benefit, and exposes active-set and execution limitations before unsupported continuation claims are made.

Appendix A

Stage A Reproducibility Appendix

A.1 Scope

This appendix records the committed Stage A evidence needed to reproduce the paper-matched Molnar candidate-v2 benchmark review without launching a new Abaqus or PBS job. It is a provenance package, not a Gate A3 pass.

A.2 Repository Revisions

Table A.1: Relevant committed revisions.

Purpose	Revision
Current supervisor package	a7294b30851d7190ee91c8bf92390bf32b2442c8
Candidate-v2 generation, static validation, and baseline submission	711dd495bdc830d695f9d7e56283316c9d417d5
SDV15 diagnostic r2 preparation and submission	209ad325d2c85532411c13d8290db08ca35b0637

A.3 Job Records

Table A.2: Stage A HPC jobs used in the paper-matched candidate-v2 evidence set.

Job	Submitted revision	Result
1374864.mmast02	711dd495bdc830d695f9d7e56283316c9d417d5	1374864.mmast02 candidate-v2 baseline. PBS Exit_status=0; Abaqus return code zero; classification paper_matched_v2_technical_pass.
1375020.mmast02	efd5f60ebb9cc6ea8ced9b508a6e0c1f1834561	1375020.mmast02 SDV15 diagnostic attempt. Failed before Abaqus due to missing git in batch PATH; no solver result.

Job	Submitted revision	Result
1375028.mmast02	209ad325d2c85532411d138290db08ca35b0637	SDV15 diagnostic r2. Abaqus technical pass; wrapper post-processing failed under cluster Python compatibility; completed-increment replay and severity audit were performed without a new solution run.

A.4 Software and Runtime Environment

The candidate-v2 baseline used modules `gcc/11.4.0`, `intel/2024.2.0`, and `abaqus/2023` on the TU Freiberg HPC system. It ran serially with one CPU. The diagnostic r2 replay provenance records Abaqus Python as 2.7.15 with GCC 8.2.1.

Table A.3: Resource records.

Quantity	Baseline job	Diagnostic r2 job
PBS job ID	1374864.mmast02	1375028.mmast02
Queue	entry_imfdkfq re-quest	normal_imfdkfq initial state
CPU request	1 CPU	1 CPU
Memory request	32gb	32gb
Walltime request	24:00:00	16:00:00
Used walltime	00:38:38	00:42:27
Used CPU time	00:35:52	not retained in run summary
Used memory	2970760kb; vmem 3565868kb; Abaqus peak about 2 GB	not retained in run summary
Execution host	mnode099/0	mnode100/0

A.5 Deck and Source Hashes

Table A.4: Candidate-v2 source and deck hashes.

Artifact	SHA-256
models/generated/molnar_gravouil_2017/paper_matched_single_notch_v2/paper_matched_single_notch_v2.inp	f4d135d6c12d42a94c1874a6453a8865b4806a4e2d3b5018141471ba9245e
models/baseline_original/molnar_gravouil_2017/02_Single_Notch_Tension/SingleNotch.for	18944e5bb2a3b7973fd0d4bff03f8e078eef667965343d8a29156d093f53f

Artifact	SHA-256
models/generated/molnar_gravouil_2017/paper_matched_single_notch_v2/SingleNotch_v2.for	e587c195c50a5c52a000ca54d3ed00e1bf8161ba2f6aa7b3cf5aacec84259

A.6 Static Validation and Diagnostic Targets

Candidate v2 static validation is recorded as `static_validation_pass` with runnable status `true`. Mesh-quality preflight is recorded as `mesh_quality_preflight_pass`. The diagnostic r2 variant used 72 target physical elements and 152 target element/IP pairs. Its deck comparison is `scientific_keyword_equivalence_pass`; its static validation is `diagnostic_static_validation_pass`; its runnable flag is `diagnostic_runnable: true`.

A.7 Trace Provenance

The SDV15 diagnostic r2 raw trace is intentionally not stored in Git. It is retained in the HPC evidence area:

```
/home/pr21vyci/projects/adaptive-remeshing/runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/evidence/1375028.mm
master02/molnar_v2_sdv15_diagnostic_call_trace.csv
```

The trace size is 258449576 bytes, modified at 2026-07-16T13:24:09+0200, with SHA-256 5b4922884ea6ee7af759c469f69e508106aca8e6d14892520abaffe2120dbed3. The completed-increment replay processed 627304 trace rows, retained 209152 U1 stage-101 rows, formed 101840 final increment states, and monitored 152 element/IP sequences.

A.8 Postprocessing Selection Rule and Tolerances

The completed-increment replay keeps the last U1 stage-101 call for every (KSTEP, KINC, physical element, source). Retained-frame events are aligned to trace increments by step time/load level, not raw retained frame number.

The replay decrease tolerance is 10^{-8} . The severity-audit materiality tolerance is 10^{-6} . The Fig. 7 digitization uncertainty is recorded as ± 0.00004 mm in displacement and ± 0.015 kN in reaction force. These are evidence definitions, not final supervisor-approved thesis tolerances.

A.9 Committed Evidence Paths

- Baseline run manifest: `runs/hpc/paper_matched_single_notch_v2/RUN_MANIFEST.md`
- Baseline run summary: `runs/hpc/paper_matched_single_notch_v2/RUN_SUMMARY.md`
- Technical evidence: `runs/hpc/paper_matched_single_notch_v2/evidence/`
- Extracted RF-U and contour states: `runs/hpc/paper_matched_single_notch_v2/extracted/`
- Scientific decision: `runs/hpc/paper_matched_single_notch_v2/scientific_review/SCIENTIFIC_DECISION.md`
- Fig. 7 comparison metrics: `runs/hpc/paper_matched_single_notch_v2/scientific_review/fig7_comparison_metrics.json`

- Crack threshold metrics: `runs/hpc/paper_matched_single_notch_v2/scientific_review/crack_path_threshold_metrics.csv`
- SDV irreversibility metrics: `runs/hpc/paper_matched_single_notch_v2/scientific_review/sdv_irreversibility_metrics.json`
- Diagnostic r2 summary: `runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/RUN_SUMMARY.md`
- Completed-increment replay evidence: `runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/evidence/1375028.mmaste02/postprocessing_completed_increment_replay_time_aligned/`
- Severity audit: `runs/hpc/paper_matched_single_notch_v2_sdv15_diagnostic_r2/evidence/1375028.mmaste02/postprocessing_completed_increment_replay_time_aligned/severity_audit/`

A.10 Excluded Files

The Abaqus ODB and raw diagnostic trace are intentionally excluded from Git. The candidate-v2 ODB remains on scratch at:

```
/scratch/pr21vyai/adaptive-remeshing/runs/molnar_paper_matched_single_notch_v2_1374864.mmaste02/molnar_paper_matched_single_notch_v2.odb
```

The raw diagnostic trace remains in the HPC evidence area and is referenced by hash in this appendix. LaTeX build products such as `.aux`, `.fdb_latexmk`, `.fls`, `.out`, generated PDFs, and `.synctex.gz` are local build artifacts.

A.11 Molnar $lc = 0.015$ mm h-convergence provenance

Table A.5: Key jobs and revisions for the supplementary $lc015$ h-convergence study.

Item	Value
Scientific-input revision	58d7e3102d76fe0e70e6729457e2c7e90ad131bb
CAE infrastructure revision	bd09bc4f33a1415bba70769458d5bbbf218e1592
H0 solver	1376154.mmaste02 (technical pass)
H1 solver	1376185.mmaste02 (technical pass)
H2-PUB solver	1376186.mmaste02 (technical pass)
Consolidated CAE	1376236.mmaste02 (all three RF-U packages)
Scientific review	runs/hpc/molnar_lc015_h_convergence/comparison/H_CONVERGENCE_SCIENTIFIC_REVIEW.md

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